CAPSTONE PROJECT REGISTRATION FORM – GFA26SE43

AI-powered Enterprise Knowledge Platform

Version 2.0 – proposed final version for Review 2 (22/10/2026). Changes from version 1.0 are listed in Form BM03.

English Title *

AI-powered Enterprise Knowledge Platform with Human-in-the-Loop Governance and Permission-aware Retrieval

Vietnamese Title *

Nền tảng quản lý và khai thác tri thức doanh nghiệp ứng dụng AI với quản trị tri thức có sự tham gia của con người và truy xuất theo phân quyền

Context *

Knowledge is one of the most valuable assets of modern enterprises. However, organizational knowledge is often scattered across documents, emails, shared folders, project repositories, and personal notes, making information retrieval inefficient and reducing employee productivity. Existing AI assistants can answer questions from enterprise documents, but many organizations hesitate to adopt them due to concerns about knowledge quality, access control, data security, and AI hallucination.

In practice, enterprise knowledge should follow a governance process in which documents are created by responsible departments, then reviewed and approved by designated knowledge managers before becoming official organizational knowledge. At the same time, employees maintain personal knowledge such as notes, technical references, meeting minutes, and learning materials, which should remain private while still being searchable by their own AI assistant.

This project proposes an AI-powered Enterprise Knowledge Platform that combines enterprise knowledge management, Human-in-the-Loop knowledge governance, permission-aware retrieval, and AI-assisted knowledge services. The platform enables organizations to manage the complete knowledge lifecycle while providing trustworthy AI-generated answers with source citation and role-based access control.

Unlike conventional enterprise chatbots, the proposed platform emphasizes knowledge governance, Human-in-the-Loop validation, permission-aware retrieval, and trustworthy AI services. The platform is designed to improve both knowledge quality and employee productivity while ensuring enterprise security and compliance.

Proposed Solutions *

The proposed system consists of five integrated modules:

1. User and Access Management

- Authentication (email/password, Google), organization and membership management with email invitations, departments, and audience-based access control with reviewer grants.

2. Knowledge Repository

- Enterprise Knowledge Repository.
- Personal Knowledge Repository.

3. Knowledge Governance

- Document submission.

- Human review by authorized reviewers (Department or Organization reviewer grants) or the Organization Admin; the approved snapshot is published automatically after it is indexed.
- Publishing and archival.
- Knowledge lifecycle management.

4. AI Knowledge Assistant

- Permission-aware Retrieval-Augmented Generation (RAG).
- AI question answering.
- AI document summarization.
- Grounded responses with source citation.

5. Dashboard and Analytics

- Knowledge usage analytics.
- AI usage statistics.
- Knowledge quality monitoring.
- User feedback management.

The platform adopts a Human-in-the-Loop approach in which AI assists knowledge retrieval while humans remain responsible for validating, approving, publishing, and continuously improving organizational knowledge.

The proposed solution manages the complete knowledge lifecycle, including document creation, submission, human review, approval, publication, retrieval, continuous feedback, and knowledge improvement. AI is designed to assist employees rather than replace human decision-makers.

Functional Requirements *

User & Access Management

- User authentication and authorization.
- Organization management: any verified user can create an organization (one installation serves several isolated organizations) and invite members by email.
- Access management: fixed capabilities (Member, Organization Admin, Arden System Admin), audience groups that define who can read published knowledge, and reviewer grants.
- Department management.
- User profile management.

Enterprise Knowledge Management

- Create enterprise documents in a rich-text editor or upload PDF files (maximum 5 MB).
- Assign metadata (scope, tags) and reader audiences; the responsible department is the author's primary department.
- Submit documents for review.
- Version tracking.

Knowledge Governance

- Review, approve, or reject submitted documents.
- Reject with a reason; the author revises the working draft and resubmits a new snapshot.
- Publish automatically after approval and indexing; unpublish (archive) documents.
- Restore archived documents.

Personal Knowledge Management

- Upload personal documents.

- Organize personal knowledge collections.
- Search and manage private knowledge.
- Optionally submit selected personal knowledge for enterprise publication through the governance workflow as a separate copy; the private original stays private.

AI Knowledge Assistant

- AI-powered semantic search.
- Enterprise question answering.
- Personal AI assistant.
- Hybrid retrieval from enterprise and personal knowledge (within user permissions).
- AI document summarization.
- Source citation for every AI response.
- Collect user feedback on AI responses.
- Allow users to rate AI-generated answers.
- Record AI interaction history.

Dashboard & Analytics

- Knowledge repository statistics (documents by status and department).
- AI usage analytics (questions per day, answered questions, average answer rating).
- Knowledge contribution reports (submissions and review decisions per department).
- User feedback dashboard (list of rated AI answers with comments).
- Knowledge quality monitoring (low-rated answers and the documents they cite).

Non-Functional Requirements *

- Secure authentication and server-side authorization (capabilities, audiences, reviewer grants) enforced by the API and PostgreSQL row-level security.
- Permission-aware retrieval to prevent unauthorized information disclosure.
- Human-in-the-Loop validation for enterprise knowledge publication.
- Human approval is required before enterprise documents become searchable by AI.
- AI services must generate grounded responses based only on authorized knowledge sources.
- AI services shall not expose unauthorized information.
- AI responses must include source citations.
- Response time below 5 seconds for common AI queries.
- Responsive web interface with RESTful APIs.
- Scalable and modular architecture.
- Docker-based deployment.
- Audit logging for all governance activities.
- Maintainable and extensible software architecture.

Theory & Practice (Document) *

Software Engineering

- Software Development Life Cycle.
- Object-Oriented Analysis and Design.
- Software Architecture.
- Database Design.

- RESTful API Development.
- Authentication and Authorization.

Artificial Intelligence

- Retrieval-Augmented Generation (RAG).
- Large Language Models.
- Semantic Search.
- Vector Databases.
- Prompt Engineering.

Knowledge Management & Trustworthy AI

- Enterprise Knowledge Management.
- Information Retrieval.
- Knowledge Governance.
- Human-in-the-Loop AI.
- Responsible AI.

Research-Based Learning

Students are expected to:

- Conduct a literature review on Enterprise Knowledge Management and Enterprise AI.
- Identify research gaps in AI-assisted enterprise knowledge systems.
- Design and implement the proposed platform.
- Conduct experiments and evaluate system performance.
- Compare the proposed solution with baseline retrieval approaches.

Products *

- Enterprise Knowledge Platform.
- Enterprise Knowledge Repository.
- Personal Knowledge Repository.
- AI Knowledge Assistant.
- Knowledge Governance Module.
- Dashboard.
- REST APIs.
- Web Application.
- Windows desktop application sharing the web user interface.

Proposed Tasks *

Phase 1 – Research & Requirement Analysis

- Literature review.
- Requirement elicitation.
- Research question formulation.
- Software architecture design.
- Database design.

Phase 2 – Core Platform Development

- User, organization and invitation management.

- Authentication and authorization.
- Enterprise knowledge repository.
- Personal knowledge repository.
- Knowledge governance workflow.

Phase 3 – AI Integration

- Document processing.
- Embedding generation.
- Vector database integration.
- Permission-aware RAG.
- AI question answering.
- AI summarization.
- Source citation.

Phase 4 – System Integration & Evaluation

- Dashboard implementation.
- Performance optimization.
- Software testing.
- Experimental evaluation.
- Performance analysis.
- User acceptance evaluation.

Phase 5 – Deployment & Documentation

- Docker deployment.
- Technical documentation.
- User manual.
- Source code finalization.
- Final report.
- Project presentation.

Expected Research Questions

- RQ1. How does Human-in-the-Loop knowledge governance improve the quality and reliability of AI-generated responses?
- RQ2. How does permission-aware retrieval improve the security of enterprise AI assistants?
- RQ3. Does combining enterprise knowledge and personal knowledge improve knowledge retrieval effectiveness?
- RQ4. Does source citation increase users' trust in AI-generated answers?

Expected Contributions

Academic Contributions

- A Human-in-the-Loop Enterprise Knowledge Platform model.
- A Permission-aware Retrieval framework for enterprise AI.
- A governance model integrating enterprise and personal knowledge.
- An experimental evaluation of trustworthy enterprise AI systems.

Practical Contributions

- Improve enterprise knowledge sharing and utilization.
- Reduce information search time.
- Enhance knowledge quality through governance workflows.
- Increase AI transparency through grounded responses and source citation.
- Provide a reusable software platform for future Enterprise AI applications.